BEING A QUASI-COMPACT IMMERSION IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- An immersion is separated, and descent of quasi-compactness and separatedness. Schemes, Lemma schemes-lemma-immersions-monomorphisms and Same chapter, Lemmas lemma-descending-property-quasi-compact and lemma-descending-property-separated, invoked at proof lines 6275-6279. Makes f quasi-compact and separated.
- The scheme-theoretic image of a quasi-compact morphism is cut out by the kernel of the map on structure sheaves. Morphisms, Lemma morphisms-lemma-quasi-compact-scheme-theoretic-image, invoked at proof lines 6280-6285. Describes the closed subscheme Z.
- Flat base change for the pushforward of the structure sheaf, and the characterisation of quasi-compact immersions. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology and Morphisms, Lemma morphisms-lemma-quasi-compact-immersion, invoked at proof lines 6286-6296. Shows the base-changed ideals agree and that X_i -> Y_i x_Y Z is an open immersion.
- Descent of open immersions. Same chapter, Lemma lemma-descending-property-open-immersion, invoked at proof lines 6296-6298. Makes X -> Z an open immersion.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L3: sources/stacks-descent/pinned-original-source.tex, source lines 6079–6079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KU: sources/stacks-descent/pinned-original-source.tex, source lines 5798–5798. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01L7: sources/stacks-chow/tags.txt, source lines 2076–2076. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a quasi-compact immersion" is fpqc local on the base.*

Proof. Let $f : X \to Y$ be a morphism of schemes. Let $\{Y_i \to Y\}$ be an fpqc covering. Write $X_i = Y_i \times_Y X$ and $f_i : X_i \to Y_i$ the base change of f. Also denote $q_i : Y_i \to Y$ the given flat morphisms. Assume each f_i is a quasi-compact immersion. By Schemes, Lemma 01L7 each f_i is separated. By Lemmas 02KQ and 02KU this implies that f is quasi-compact and separated. Let $X \to Z \to Y$ be the factorization of f through its scheme theoretic image. By Morphisms, Lemma 01R8 the closed subscheme $Z \subset Y$ is cut out by the quasi-coherent sheaf of ideals $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_Y \to f_*\mathcal{O}_X)$ as f is quasi-compact. By flat base change (see for example Cohomology of Schemes, Lemma 02KH; here we use f is separated) we see $f_{i,*}\mathcal{O}_{X_i}$ is the pullback $q_i^* f_*\mathcal{O}_X$. Hence $Y_i \times_Y Z$ is cut out by the quasi-coherent sheaf of ideals $q_i^*\mathcal{I} = \mathrm{Ker}(\mathcal{O}_{Y_i} \to f_{i,*}\mathcal{O}_{X_i})$. By Morphisms, Lemma 01RG the morphisms $X_i \to Y_i \times_Y Z$ are open immersions. Hence by Lemma 02L3 we see that $X \to Z$ is an open immersion and hence f is an immersion as desired (we already saw it was quasi-compact). □